

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“VALIDATION OF CREDIT RISK MODEL FOR A MICROFINANCE INSTITUTION”

Submitted in partial fulfillment for the award of degree (21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Yash Govil (1DS21AI057)

Conducted at

EQUIFAX

Equifax India

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**VALIDATION OF CREDIT RISK MODEL FOR A MICROFINANCE INSTITUTION**” carried out by **Yash Govil [1DS21AI057]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

**Signature of
Reporting Head**
Mr. Rami K

AVP-Data & Analytics
Equifax

Signature of Internship Guide
Dr. Reshma S
Dept. of AIML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Yash Govil**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date: 23/05/2025

Place: Bangalore

USN: 1DS21AI057

NAME: Yash Govil

OFFER LETTER



LETTER OF OFFER AS TRAINEE ("LOT")

Date: 10 February 2025

From
Human Resources Department,
Equifax Credit Information Services Pvt. Ltd ("Equifax")
Unit No. 931, 3rd Floor, Building No.9,
Solitaire Corporate Park,
Andheri Ghatkopar Link Road,
Andheri East, Mumbai – 400093, India

To
Mr Yash Govil ("You"),
S/o of Sunil Kumar Govil
Aged about 22 yrs

Residing at: C-06/02, Platinum City, HMT Road, Yeshwanthpur, Bangalore, Karnataka 560022

Ref: Regarding your letter/application dated 27/12/2025 ("Letter") expressing your interest to be provided an opportunity as a Trainee at Equifax.

Dear Mr Yash Govil,

In reference to your letter communicating interest to be provided with an opportunity as Trainee at Equifax, we would like to confirm that your application has been accepted. Here are the terms and conditions that will be applicable to You while being a Trainee at Equifax.

Equifax and You shall be collectively referred to as "Parties"

You represent to Equifax that You are pursuing Bachelor of Engineering specialization in Artificial Intelligence and Machine Learning course at Dayanand Sagar College and in pursuance of the prescribed curriculum by your Institution/University You have reached out to Equifax for an opportunity to be a Trainee at Equifax.

1. You will be designated as a 'Trainee'. You will be trained under the guidance and supervision of Sudipta Mondal (E90088178) designation AVP - Analytics ("Reporting Supervisor") at Equifax and you will at all time follow instructions and directions of your Reporting Supervisor. Your work location will be at our office in Bangalore and any other location as may be assigned by your Reporting Supervisor. Your working hours will be from 10.00 am to 6.00 pm (Monday to Friday). Seven (7) working days before the end of the Term, You shall make a detailed presentation to the Head of Department along with Reporting Supervisor relating to the project/assignment You have undertaken as a Trainee at Equifax.
2. You will be associated with us as Trainee from 18 February 2025 to 07 July 2025 ("Term"). Equifax is at liberty to alter the timeline of the Term, including terminating the same in the event Equifax is of the view that your continuation as Trainee is jeopardizing/affecting the interest of Equifax in any manner.

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3. You will receive a stipend of INR 10,000 (INR Ten Thousand) Per Month. You will not be entitled to any other perks or benefits during your tenure as Trainee.
4. During your Term as Trainee, You will strictly comply, adhere and agree to be bound by all the Equifax policies, guidelines, code of conduct and any other internal documents of Equifax.
5. You understand and acknowledge that anything and everything associated with Your project/assignment is confidential and proprietary to Equifax, during the Term, it may be necessary for Equifax to share such confidential and proprietary information (collectively "Confidential Information") with You so as to enable You to perform duties and responsibilities as a Trainee. You shall at times ensure confidentiality and safeguarding of our Confidential Information. You further understand and agree that all Intellectual Property in and relating to the project/assignment You are part of, is and shall solely belong to Equifax, this includes any and all of Your ideas, documentation, charts, inventions, discoveries or any other outcomes of similar nature during your Term as Trainee. Your obligation of confidentiality shall continue even after the end of Term.
6. You shall return all the Company content, materials, and all work product to the Company at its earliest convenience, but in no event beyond thirty (30) days after the date of termination of this Project Assignment Letter
7. Post successful completion of the training, You will be issued a certification of completion of Training by Equifax, however the same shall be subject to revocation) in the event Equifax becomes aware of any unauthorized use/access or failure on Your part to return any materials (tangible and non-tangible) associated with your project/assignment to Equifax.
8. You shall be completely responsible for any liability or damage arising out of or in connection Your acts and omissions as a Trainee with Equifax and You agree to hold harmless Equifax from any such liability or damages. You represent and warrant that the performance and obligations by You under the project/assignment at Equifax will not violate or infringe upon the rights of any third-party including but any third-party Intellectual Property.
9. Nothing in this letter shall be construed to create an employer-employee or principal-agent relationship between the Trainee and Equifax. Under no circumstances Equifax shall be liable for any direct, indirect, special, incidental or consequential damages or liability arising out of or in connection with this LOT as Trainee.
10. The Parties agree that any dispute or issue arising out of or in connection with this LOT as Trainee shall be governed by Indian Laws and the courts at Mumbai shall have exclusive jurisdiction.

You are hereby requested to read, review and understand the contents of this LOT as Trainee carefully before signing. Once signed, kindly return a copy of this letter by **13 February 2025** to Equifax at the address stated herein.

For Equifax Credit Services Private Limited

Yash Govil

Signature: Mario Discuza
Mario Discuza (Feb 11, 2025 20:05:00 GMT+05:30)

Signature: [Signature]

Signed for and on behalf of Equifax

Feb 11, 2025

Date: Feb 12, 2025

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COMPLETION CERTIFICATE



April 04, 2025

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Yash Govil (E90115581)[C]** is undergoing the internship project with us and the details are as below.

Project Guide : Sudipta Mondal (E90088178)
Internship Start Date : February 18, 2025

Below are his contributions since his joining:

- Commercial ECR-Model Rebuild
- Demography and Dimensions
- Manual ERS Scoring

We wish him a successful and fulfilling future ahead.!

for Equifax Credit Information Services Pvt. Ltd.,

A handwritten signature in blue ink, appearing to be "Gayathri Mani", written over a horizontal line.

Gayathri Mani
Senior Manager – HR Business Partner

PS: The Interns are strictly advised not to use the confidential details but not limited to the client details, real time data, PII, any other data that are classified as per Equifax Policy. Any deviation will result in termination of internship and strict action will be taken basis the severity of the issue.

Registered Office:
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ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Rami K**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Guide, **Dr. Reshma S**, Associate Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Yash Govil [1DS21AI057]

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Effective credit risk assessment is essential for ensuring financial stability and making informed lending decisions. This study focuses on validating the predictive accuracy and stability of an existing credit risk model across three borrower segments: New To Institution (NTI), New To Credit (NTC), and Existing To Credit (ETC). The validation process employs key statistical metrics, including the Population Stability Index (PSI) to assess population shifts, the Kolmogorov-Smirnov (KS) statistic to evaluate the model's discriminatory power, and performance measures such as Area Under the Curve (AUC) and the Gini coefficient to determine predictive effectiveness. Additionally, rank order stability was analyzed to identify any breaks in risk differentiation, while the Characteristic Stability Index (CSI) was used to examine variations in key predictive variables over time. The findings reveal that NTI exhibits the highest level of instability, with significant shifts in borrower characteristics that could impact model performance. NTC demonstrates relatively stronger predictive capabilities, while ETC shows signs of model degradation, necessitating further investigation. The analysis also highlights key variables that have undergone significant distributional shifts, suggesting potential model drift. These insights underscore the importance of continuous monitoring and recalibration to maintain model accuracy. By leveraging these findings, financial institutions can refine their credit risk models, enhance risk segmentation, and optimize lending strategies to ensure a more robust and data-driven approach to credit assessment. Ultimately, this study contributes to strengthening credit risk management frameworks, improving loan portfolio performance, and fostering responsible lending practices.

1.2 Overview of the Internship

This internship focused on credit risk modelling and validation, a crucial aspect of financial risk management. Credit risk refers to the potential loss that a lender may incur if a borrower fails to meet their debt obligations. To mitigate this risk, financial institutions use credit scoring models that assess the likelihood of default based on historical data and borrower characteristics.

The internship involved working with microfinance institution (MFI) data, where credit risk modelling plays a key role in determining lending decisions. A pre-existing model was already deployed, and the primary objective was to validate its effectiveness for the current borrower population. Model validation ensures that the risk assessment remains accurate and stable over time, despite changes in borrower behaviour and economic conditions. The project provided hands-on experience with real-world credit risk data, statistical analysis, and model validation techniques, contributing to a deeper understanding of risk assessment in the financial sector.

1.3 Objectives of the Internship

- **Understand Credit Risk Modelling & Validation** – Gain in-depth knowledge of credit risk assessment, scoring models, and validation techniques used in financial institutions.
- **Analyse Microfinance Institution (MFI) Data** – Perform exploratory data analysis (EDA) on MFI data to understand borrower trends and credit-seeking behaviour.
- **Validate an Existing Credit Risk Model** – Assess the performance and stability of a pre-deployed credit scoring model using statistical techniques and validation metrics.
- **Evaluate Model Stability & Performance** – Use key metrics such as Population Stability Index (PSI), Characteristic Stability Index (CSI), AUC, Gini, and KS to measure the model's effectiveness over time.
- **Gain Hands-on Experience with Data Science Tools** – Work with Python, SAS, and BigQuery for data handling, statistical analysis, and model validation in a real-world financial setting.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Equifax India Ltd



2.1 Introduction

Equifax India is a leading credit information company that provides comprehensive credit risk assessment solutions to financial institutions, microfinance lenders, and other credit providers. As a subsidiary of Equifax Inc., a global data analytics and technology company headquartered in the United States, Equifax India plays a crucial role in the country's financial ecosystem by offering advanced credit bureau services, analytics, and decisioning solutions. The company collects and analyses vast amounts of consumer and commercial credit data to help lenders make informed decisions, reduce risk, and promote responsible lending practices. Equifax India serves a wide range of financial institutions, including banks, non-banking financial companies (NBFCs), and microfinance institutions (MFIs), enabling them to assess borrower creditworthiness, detect fraud, and improve loan approval processes.

The organization is known for its robust data security measures, strict adherence to regulatory frameworks such as RBI guidelines, and continuous innovation in credit scoring methodologies. Equifax India also provides value-added services such as portfolio monitoring, credit trend analysis, early warning systems, risk modelling, and customized insights to support financial institutions in optimizing their credit strategies. These services are designed not only to strengthen underwriting capabilities but also to enhance post-lending monitoring and portfolio health. Through its extensive database and cutting-edge analytics capabilities, Equifax India contributes significantly to financial inclusion by helping lenders extend credit to underserved and first-time borrowers while ensuring responsible borrowing and lending practices that foster trust and sustainability in the credit market.

2.2 Department/Team Overview

The Data and Analytics department at Equifax India plays a vital role in developing advanced data-driven solutions to support financial institutions in making informed credit decisions. This department is responsible for building predictive models, including the widely adopted Equifax Risk Score, which helps lenders assess borrower creditworthiness, forecast defaults, and manage risk effectively. The team leverages state-of-the-art machine learning algorithms, statistical modelling, and big data processing techniques to develop and refine models suited to dynamic market conditions.

In addition to model development, the department works on multiple strategic initiatives such as segment-specific risk analytics, product performance tracking, and credit behavior research. These efforts are aimed at creating tailored decisioning solutions that address unique challenges faced by different lender segments, including rural MFIs and urban NBFCs. The department collaborates closely with both internal product teams and external clients to deliver actionable insights that support real-time decision-making and long-term credit strategy. With a strong emphasis on innovation, compliance, and data accuracy, the Data and Analytics team ensures that Equifax India remains at the forefront of credit risk management, regulatory technology (RegTech), and decisioning solutions in an increasingly digitized financial environment.

CHAPTER 3

SCOPE OF WORK

3.1 Project/Task Overview

This project focuses on the validation of an already deployed credit risk model to assess its effectiveness in predicting creditworthiness for the current population. Model validation is a critical process to ensure that a model continues to perform accurately over time, considering potential changes in borrower behavior, economic conditions, and lending patterns. The primary objective is to evaluate whether the model remains reliable in assessing risk for microfinance institution (MFI) borrowers. The validation process includes statistical assessments, stability checks, and performance evaluations to determine if the model aligns with the current credit landscape. If discrepancies are identified, necessary adjustments or recalibrations may be recommended to maintain predictive accuracy.

Before conducting the validation, an in-depth understanding of credit risk modeling and its components was essential. The process began with a study of the history of credit risk modeling, followed by roll rate analysis, which helps track how delinquent accounts progress over time. Next, Weight of Evidence (WOE) and Information Value (IV) were explored to assess the predictive power of different variables. Logistic regression, the core methodology behind the deployed model, was studied in detail, including its mathematical formulation and significance in credit risk prediction. Further, key performance evaluation techniques such as the KAS table, Population Stability Index (PSI), Characteristic Stability Index (CSI), and Gini coefficient were analyzed to understand their role in assessing model stability and discriminatory power.

Since the model is specifically designed for MFI data, an exploratory data analysis (EDA) was conducted on the dataset to understand borrower demographics, loan characteristics, and repayment patterns. This step was crucial in identifying data distributions, trends, and potential biases within the dataset. Once the preliminary analysis was completed, the actual validation process commenced, where model outputs were compared against real-world data to check for deviations in performance. Stability tests were conducted using PSI and CSI to determine whether the model's predictions were consistent across different time periods. Additionally, the model's ability to rank-order risk was measured using the Gini coefficient. The findings from this validation process play a crucial role in deciding whether the model

remains effective or requires recalibration to ensure optimal performance in the evolving credit market.

Methodology & Approach

The model validation process followed a structured approach to assess whether the deployed credit risk model remained effective for the current population. The methodology included descriptive statistics, waterfall analysis, roll rate analysis, stability checks, and performance evaluation metrics. The data provided by the MFI client covered a period from November 2021 to December 2022, and the analysis was conducted on both application variables and categorical variables to gain insights into borrower characteristics and model performance.

Descriptive Statistics

The initial step involved conducting descriptive statistics on the dataset provided by the MFI client. This helped in understanding the distribution of key variables, missing values, and overall data quality. The dataset consisted of two types of variables:

- **Application variables:** Related to borrower attributes, such as age, income, and employment details.
- **Categorical variables:** Representing grouped borrower characteristics, such as loan type and tenure categories.

By performing descriptive statistics, a preliminary understanding of the dataset was established, allowing for segmentation and further analysis.

The dataset consists of a total of 872,064 applications recorded over the observation period from November 2021 to December 2022. The application variables exhibit complete data availability except for Var_9, which has 1,033 missing values. The categorical variables provide insights into the applicant distribution across different categories. State-wise distribution shows that State_1 has the highest representation (27.81%), followed by State_2 (18.54%), while State_5 has the lowest share (8.02%). In terms of house ownership, the majority of applicants (98.02%) own their houses, while only 1.98% are in rental accommodations. Home area distribution indicates that most applicants reside in rural areas (71.53%), with smaller proportions in semi-urban (19.18%), urban (8.99%), and sub-urban (0.30%) regions. Business ownership statistics reveal that 98.72% of applicants are business owners, while a small fraction (1.28%) do not own a business, with further breakdowns

indicating minor variations in responses. These descriptive statistics provide an overview of the dataset composition and highlight key demographic and financial characteristics of the applicants.

This preliminary descriptive analysis serves as a foundation for further stability checks (PSI, CSI), predictive power evaluations (Gini coefficient), and roll rate analysis.

Waterfall Analysis and Segmentation

A waterfall analysis was conducted to track borrower movements across different segments and assess the distribution of good and bad borrowers. The segmentation included:

- **Existing to Credit (ETC):** Borrowers who had a prior credit history.
- **New to Institution (NTI):** Borrowers who were new to the lending institution but had previous credit history elsewhere.
- **New to Credit (NTC):** Borrowers with no prior credit history

Each segment was analyzed separately to determine differences in credit risk behavior. The bad definition (BD) was set at 60+ days past due (DPD), meaning borrowers with overdue payments beyond 60 days were categorized as bad.

Variable-wise Binning and Population Analysis

To further evaluate model performance, a variable-wise binning approach was adopted. Within each segment, key variables were binned, and their respective scores and population distributions were analyzed. This step helped in:

- Understanding how different borrower attributes contributed to risk prediction.
- Identifying variables with strong discriminatory power.
- Assessing population stability across different time periods.

By analyzing the distribution of score within each segment, potential deviations or anomalies in model predictions were identified.

CHAPTER 4

LEARNING OBJECTIVES

1. Develop Proficiency in Credit Risk Modelling and Validation

The primary learning objective of the internship was to acquire in-depth understanding of credit risk modelling and validation techniques. The student learned how financial institutions assess borrower creditworthiness using historical data, and how logistic regression models are used to predict loan default risk. Through the project, the intern gained expertise in critical metrics such as Population Stability Index (PSI), Characteristic Stability Index (CSI), Kolmogorov-Smirnov (KS) statistic, AUC, and the Gini coefficient. This objective enabled the intern to evaluate the effectiveness and stability of risk models over time and understand the importance of maintaining model relevance amid changing borrower behavior.

2. Perform Data Analysis Using Industry Tools and Large-Scale Datasets

This project aimed to build proficiency in using real-world data science tools such as BigQuery, Python, and SAS for data handling and statistical evaluation. The intern worked with large volumes of Microfinance Institution (MFI) data to perform exploratory data analysis (EDA), identify borrower trends, and detect population shifts. BigQuery facilitated fast data retrieval and aggregation, while Python aided in visualization and metric computation. SAS was employed for model validation and performance testing. This objective emphasized practical experience in managing messy datasets, executing advanced queries, and leveraging statistical tools for insights into credit risk dynamics.

3. Gain Hands-On Experience with Industry-Standard Tools for Data Handling and Model Validation

A significant learning objective of the project was to develop hands-on experience with industry-relevant tools such as **Python**, **SAS**, and **Google BigQuery** to handle real-world credit data and perform model validation. The intern used **BigQuery** for querying and managing large-scale datasets efficiently, ensuring structured and accessible data pipelines. **Python** was used extensively for data preprocessing, exploratory data analysis (EDA), visualization, and computing metrics like PSI and CSI. **SAS** enabled statistical validation through performance metrics like AUC, KS, and Gini coefficient. Mastery of these tools helped bridge theoretical knowledge with practical applications in credit risk analytics and

financial data science.

4. Conduct Comprehensive Model Stability and Performance Evaluation

A core learning outcome was the ability to assess the ongoing effectiveness of an existing credit risk model using stability and performance metrics. The intern calculated PSI and CSI to identify population and variable-level shifts, respectively. These insights helped determine if the model experienced performance degradation or drift over time. Additionally, by evaluating KS, AUC, Gini, and detecting rank order breaks, the intern gauged the model's discriminatory power across borrower segments (NTC, NTI, ETC). This process taught how to ensure model consistency, recommend recalibrations, and confirm the continued utility of the model for credit risk evaluation.

5. Enhance Problem-Solving and Reporting Skills in Real-World Projects

A key outcome was the development of analytical thinking and effective communication skills while working on a real-world project. The intern encountered challenges such as inconsistent data, high instability in borrower characteristics, and rank order breaks. These were addressed through methodical problem-solving—data cleaning, roll rate validation, and model recalibration. The intern also learned to synthesize findings into clear, impactful reports using visualizations and structured insights. By translating technical analysis into business-friendly recommendations, the intern gained experience in bridging data science with strategic decision-making—critical for careers in financial analytics, risk consulting, or FinTech domains.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Inconsistent Reporting Standards Across MFIs

Challenges:

During the validation of the credit risk model, data was sourced from multiple Microfinance Institutions (MFIs), each following its own reporting standards and formats. This inconsistency led to difficulty in aligning key variables such as income range, loan tenure, and repayment behavior. Mismatched column names, categorical labels, and missing values resulted in delays and risked compromising the accuracy of the validation process.

Solutions:

A standardized data dictionary was developed, and transformation scripts were written in Python to unify variable formats across all datasets. This normalization ensured consistency in analysis, reduced preprocessing time, and improved the quality and reliability of statistical validation outputs.

2. PSI and CSI Misinterpretation Due to Improper Binning

Challenges:

Initially, the computation of Population Stability Index (PSI) and Characteristic Stability Index (CSI) produced skewed or misleading results due to poorly defined binning strategies. Arbitrary bin ranges caused irregular distributions and failed to capture genuine shifts in population behavior, making the model appear less stable than it was.

Solutions:

Custom binning logic was implemented using quantile-based segmentation in Python to ensure even distribution across bins. Additionally, domain-specific thresholds were integrated for key variables. This approach led to more accurate PSI/CSI outputs, enabling better detection of meaningful data drifts and model degradation signals.

3. Low Performance Metrics for ETC Segment

Challenges:

While validating the model across borrower segments, the Existing to Credit (ETC) group consistently showed lower performance metrics (e.g., KS statistic below 15 and AUC under 0.60). This suggested weak separation between good and bad borrowers and raised concerns

about the model's relevance for experienced borrowers.

Solutions:

A focused exploratory analysis was conducted to identify underperforming features in the ETC segment. Based on the insights, feature selection and score recalibration techniques were proposed. These adjustments improved the segment's model performance and restored its ability to effectively differentiate borrower risk.

4. Manual Validation Metric Calculation Led to Errors

Challenges:

During initial evaluations, key model performance metrics like KS, AUC, and Gini were calculated manually using spreadsheets or ad hoc scripts. This process was time-consuming and prone to human error, particularly when working with Out-of-Time (OOT) datasets or comparing across borrower segments.

Solutions:

Reusable Python and SAS scripts were developed to automate the computation and visualization of validation metrics. These scripts included built-in validation checks and automated report generation. As a result, the validation process became faster, error-free, and scalable for future model monitoring cycles.

5. Rank Order Breaks Affecting Risk Score Credibility

Challenges:

The model exhibited rank order breaks in several score buckets, where lower-scoring borrowers were performing better than higher-scoring ones. This undermined the credibility of the risk score and raised red flags for potential model instability, especially in segments like NTI (New to Institution).

Solutions:

Detailed score band analysis was performed, and monotonicity constraints were reviewed. Data transformation techniques and logistic model coefficients were adjusted to restore the proper risk order. Post-correction, the scorecard regained its logical structure, improving both stakeholder confidence and operational usability.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

Operating System: Windows 10/11

Development Tools:

- Jupyter Notebook / JupyterLab
- SAS Enterprise Guide
- Google Cloud SDK

Programming Languages and Frameworks:

Data Processing:

- Language: Python
- Libraries Used:
 - Pandas – for data manipulation and analysis
 - NumPy – for numerical computing
 - Matplotlib / Seaborn – for data visualization
 - Scikit-learn – for model evaluation (AUC, Gini, KS)
 - Scipy – for statistical calculations
 - Statsmodels – for logistic regression and statistical summaries

Database:

- Google BigQuery – for storing and querying large-scale MFI datasets
- CSV and Excel – used as intermediate data files

Hardware requirements:

- Processor: Intel i5 or AMD Ryzen 5 and above
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 256 GB SSD
- Internet: Stable high-speed connection

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

6.1. Key Deliverables

6.1.1. Data Quality Assessment

- Verified the integrity of MFI data for the period November 2021 – December 2022.
- Ensured data consistency, completeness, and accuracy before proceeding with validation.
- Identified shifts in borrower profiles that could impact model performance.

6.1.2. Bad Definition Confirmation

- Evaluated delinquency patterns using roll rate analysis.
- Validated the definition of bad borrowers as **60+ DPD (Days Past Due)**.
- Ensured that the bad definition remained relevant for identifying high-risk borrowers.

6.1.3. Model Stability Analysis

• Population Stability Index (PSI):

- Assessed whether the borrower population had changed significantly over time.
- Identified score drift and potential model degradation.

• Characteristic Stability Index (CSI):

- Measured changes in key predictive variables.
- Recommended recalibration for variables with significant shifts.

6.1.4. Segment-Wise Model Performance Evaluation

- Analyzed model behavior across three borrower segments: **ETC, NTI, and NTC**, using waterfall analysis.
- Ensured that the model effectively differentiated risk across all segments.

6.1.5. Model Predictive Power Evaluation

- **AUC (Area Under the Curve):** Evaluated the model's ability to classify borrowers correctly.
- **Gini Coefficient:** Assessed the predictive strength of the score.
- **KS Statistic (Kolmogorov–Smirnov Test):** Verified how well the model separated good and bad borrowers.

6.1.6. Final Recommendations & Next Steps

- Provided a validation report summarizing:
 - Model stability, variable shifts, and risk differentiation effectiveness.

- Areas requiring recalibration or adjustments.
- Suggested continuous monitoring of model performance to track evolving borrower behavior.
- Recommended enhancements in segmentation, variable transformations, and score thresholds where needed.
- Ensured that the model remains **reliable, stable, and relevant** for predicting credit risk in the current borrower population.

6.2. Impact of Work

The validation of the deployed credit risk model had a significant impact on the accuracy, reliability, and decision-making process within the microfinance sector. By thoroughly assessing the model's stability and predictive power, the validation ensured that lending institutions could continue using the score effectively to differentiate between high-risk and low-risk borrowers. One of the most critical aspects was identifying population shifts and changes in borrower behavior, which could potentially lead to model degradation over time. Through Population Stability Index (PSI) and Characteristic Stability Index (CSI) calculations, it was determined whether the model required recalibration due to changes in key features or borrower demographics. This validation process directly contributed to maintaining fair and responsible lending practices, ensuring that borrowers were assigned credit scores based on the most current and relevant risk indicators.

Additionally, the segmentation of borrowers into Existing to Credit (ETC), New to Institution (NTI), and New to Credit (NTC) played a vital role in refining credit risk assessment strategies. By analyzing the roll rates, delinquency patterns, and score distribution within each segment, the validation process helped lenders understand which borrower groups posed higher risks and required closer monitoring. The assessment of rank order stability and risk differentiation metrics such as AUC, Gini, and KS ensured that the model was still performing optimally across different borrower segments. This helped in optimizing lending policies, refining risk-based pricing strategies, and improving portfolio risk management. Institutions were able to make data-driven decisions, ensuring that creditworthy borrowers continued to have access to financing while reducing exposure to high-risk applicants.

Beyond model performance, the validation work also contributed to regulatory compliance and risk governance. Financial institutions and regulatory bodies require rigorous validation

and monitoring of risk models to ensure they align with industry standards. By confirming the stability and predictive power of the model, the validation process provided lenders with the confidence that their credit risk assessment frameworks remained robust and compliant. Furthermore, the insights from the validation allowed institutions to implement proactive measures such as periodic recalibration, enhanced data monitoring, and better feature selection for future models. Ultimately, the impact of this validation work extended beyond immediate model adjustments—it laid the foundation for continuous improvement in credit risk modeling, enhanced financial inclusion, and a more resilient lending ecosystem for microfinance institutions.

CHAPTER 7

REFLECTION AND ANALYSIS

7.1 Summary of Experience

The internship at Equifax India in the Data and Analytics department provided a comprehensive exposure to credit risk modelling, model validation, and risk assessment methodologies. The primary focus was on validating an already deployed credit risk model, ensuring its stability and effectiveness in predicting borrower risk for the current population. The project required an in-depth understanding of credit risk concepts, including roll rate analysis, Weight of Evidence (WOE) transformation, logistic regression, and stability indices like PSI and CSI. Additionally, exploratory data analysis (EDA) was conducted on Microfinance Institution (MFI) data to gain insights into borrower behavior and trends over a specified period.

Beyond technical aspects, the experience offered valuable insights into the operational workflows of a financial bureau, the importance of model monitoring and governance, and the impact of credit scoring on financial decision-making. Working on real-world financial data and models enhanced problem-solving skills, analytical thinking, and adaptability to large-scale data challenges. The structured approach to model validation—covering descriptive statistics, waterfall analysis, segmentation, stability testing, and predictive power assessment—reinforced the critical role of data-driven decision-making in financial risk management.

7.2 Skills Acquired

7.2.1. Technical Skills

- **Credit Risk Modelling:** Gained expertise in roll rate analysis, logistic regression, and WOE transformation for risk assessment.
- **Model Validation Techniques:** Learned to evaluate model performance using PSI, CSI, AUC, Gini Coefficient, KS Statistic, and rank order stability.
- **Exploratory Data Analysis (EDA):** Conducted descriptive statistics, segmentation analysis, and data quality checks.
- **BigQuery for Data Handling:** Worked with datasets processed using BigQuery for

validation and statistical assessments.

- Python for Data Analysis: Used Python for EDA, feature analysis, and visualization of model performance metrics.
- SAS for Statistical Modelling: Utilized SAS for advanced statistical analysis and credit risk modelling techniques.
- Waterfall Analysis: Applied segmentation techniques (ETC, NTI, NTC) to understand borrower risk categories.
- Scorecard Evaluation: Assessed variable-wise binning, monotonicity checks, and score distribution consistency.

CHAPTER 8

RECOMMENDATIONS

1. Introduce Well-Defined Internship Structures

Equifax can enhance future internship experiences by adopting a structured framework that sets clear expectations from the start. Defining roles, goals, timelines, and deliverables will help interns stay focused and aligned. A balanced schedule with interactive sessions and task-based learning can guide skill growth while enabling interns to track their development more effectively.

2. Foster Academic Collaborations

Establishing partnerships with universities and institutions can enrich internships by connecting theory with practical applications. Collaborations through workshops, guest lectures, or joint research activities can introduce interns to real-world challenges in cybersecurity, risk management, and compliance, making their learning more well-rounded.

3. Establish Strong Mentorship Channels

Assigning each intern a mentor who offers technical guidance and regular feedback can make the learning curve smoother. Structured mentor–intern check-ins, feedback reviews, and support can help interns better understand workflows and boost their confidence in applying tools and solving problems independently.

4. Encourage Active Participation and Creative Problem Solving

Creating opportunities for interns to contribute ideas, participate in meetings, and take ownership of smaller projects encourages innovation. This involvement not only enhances their learning but also brings fresh ideas to ongoing work, fostering a two-way benefit between the intern and the organization.

5. Create Networking and Exposure Opportunities

Facilitating cross-functional interactions can help interns explore various career paths. Organizing Q&A sessions, informal chats with professionals, or internal demo days allows interns to learn beyond their immediate teams, increasing awareness of the broader tech and cybersecurity landscape.

CHAPTER 9

CONCLUSION

My internship journey at Equifax has been an incredibly enriching and transformative experience. Over the course of this internship, I not only gained technical knowledge but also developed a deeper understanding of how real-world projects work in a professional environment. The successful completion of this project provided an in-depth understanding of the principles, practices, and challenges associated with validating a credit risk model in the microfinance domain. The primary objective was to assess the stability and predictive performance of an already deployed credit scoring model used to evaluate borrowers from different segments—New to Credit (NTC), New to Institution (NTI), and Existing to Credit (ETC). This was accomplished through rigorous statistical methods including Population Stability Index (PSI), Characteristic Stability Index (CSI), and key performance metrics like AUC, Gini Coefficient, and the Kolmogorov–Smirnov (KS) statistic.

The findings revealed critical insights into model behavior over time, highlighting population drift and rank order breaks, especially in the NTI and NTC segments. These deviations pointed toward a need for recalibration to preserve the model's effectiveness and fairness in scoring borrowers. Segment-wise analysis allowed for a more targeted understanding of risk, reinforcing the importance of customizing model parameters based on borrower profiles rather than relying on a uniform strategy.

In addition to technical knowledge, the project fostered key skills in data preprocessing, exploratory data analysis (EDA), statistical programming using Python and SAS, and working with large-scale datasets via Google BigQuery. These tools played a pivotal role in enabling efficient data manipulation, analysis, and reporting. The hands-on experience strengthened the ability to interpret complex data, detect model weaknesses, and communicate findings effectively to stakeholders.

Overall, the project underscored the dynamic nature of credit risk modeling and the critical need for continuous validation in real-world financial environments. The insights generated not only enhanced model governance but also contributed toward improving the decision-making frameworks within lending institutions. This experience has laid a strong foundation for future endeavors in the fields of risk analytics, financial modeling, and data-driven policy-making.

CHAPTER 10

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